

Noise Robust Radar HRR Target Recognition Based on Bayesian Sparse Learning

Lan Du, Penghui Wang, Lei Zhang, Hongwei Liu, Danlei Xu
National Lab. of Radar Signal Processing, Xidian University, Xi'an, China
Email: dulan@mail.xidian.edu.cn

Abstract—A noise robust statistical model based on Bayesian sparse learning (BSL) is developed to characterize the complex-valued high range resolution (HRR) radar target signal, motivated by the problem of radar automatic target recognition (RATR). We assume a sparseness-promoting prior on the complex echoes from the scattering centers and a Markov dependency for the location of the dominant scattering center between consecutive HRR signals in the hierarchical Bayesian model. Considering the low signal-to-noise ratio (SNR) problem for a test sample, the statistical model trained under the high SNR can be updated to match the measured test sample and the corresponding recognition decision can be made based on the updated model. Efficient inference is performed via variational Bayesian (VB) for the proposed Bayesian sparse model. To validate the formulation, we present the experimental results on the measured HRR dataset for target recognition and signal reconstruction, and provide comparisons to some other statistical models for RATR.

I. INTRODUCTION

Since the high range resolution (HRR) radar target signals contain abundant target information, such as target size, scatterer distribution, *etc.*, HRR radar target recognition, especially for the detailed target model recognition, has received intensive attention from the radar automatic target recognition (RATR) community [1]–[12]. Several literatures [1]–[7], [9], [10] showed that statistical recognition is an efficient method for HRR real data (*i.e.*, HRRP) based RATR. Some parametric statistical models are developed to describe the class-conditional likelihood of HRRP data, *e.g.*, Gaussian model [1], [2], Gamma model [3], the two-distribution compounded model comprising Gamma distribution and Gaussian mixture distribution [4], factor analysis (FA) model [5], local factor analysis (LFA) model [6], and the multitask learning (MTL) based hidden Markov model (HMM) [7]. In addition, HMM is also utilized to characterize the feature vector extracted from the HRRP sequence in [9], [10].

The main disadvantage of the statistical modeling for the HRRP data or the FFT-magnitude feature of the HRR complex data is that it is difficult to modify the statistical model for the real-valued features to match the test data measured under the low signal-to-noise ratio (SNR) condition. For the RATR problem, the training data are usually collected under the condition of high SNR via some cooperative measurement experiments or directly via simulations; while the test samples are usually got in the non-cooperative circumstance (*e.g.*, at the battle time), where the high SNR cannot be guaranteed due to the severe measurement conditions, such as the large

distance between the non-cooperative targets and radar. Thus the statistical model should be modified to match the test data measured under the low SNR in the classification stage to obtain the robust recognition performance. For the radar signal processing, the noise directly affects the complex-valued target echo, which can be expressed as an additive model, but the absolute value operator to obtain the real-valued feature will completely change the signal model, *i.e.*, the additive model does not hold for the noised HRRP data or the FFT-magnitude feature of the HRR complex data. Based on the additive real-valued noise assumption, [8] proposes a noise-robust modification method for MTL-FA model. Nevertheless, according to the results in [8], although the noise robustness of the modified MTL-FA model is superior to those of the conventional statistical models, the proposed algorithm still can not yield the good recognition performance when $\text{SNR} < 15\text{dB}$, since the noise model assumption is not correct.

The research reported here first seeks a signal model for HRR complex data based on Bayesian sparse learning (BSL). The statistical model for the HRR complex data from each target-aspect can be analytically derived based on the proposed signal model, which can be applied to statistical target recognition. In the experiments based on measured HRR dataset, the proposed methods show the inspiring performance on target recognition for the test samples under the low SNR condition.

II. SIGNAL MODEL BASED ON BAYESIAN SPARSE LEARNING

For an HRR radar system, an aircraft-like target does not appear as a “point target” any longer, but consists of a set of scattering centers separated in several range cells along the radar line-of-sight (LOS). An HRR target signal represents the distribution of target scattering centers in the radial range dimension. HRR scattering from a complex target yields target signature that is a strong function of the target-sensor orientation, referred to as target-aspect sensitivity in [4]–[8], [13]. To deal with the target-aspect sensitivity, the multi-aspect HRR dataset from each target is divided into subsets (as in [4], [5], [8], [13]), where the HRR data in each subset are collected from a target-aspect sector roughly without the scatterers’ motion through range cells (MTRC). Thus each subset is defined to be an aspect-frame from the target [4], [5], [8], [13].

According to [14], the high-frequency scattering response of an object is well approximated as a sum of responses from

individual scattering centers [14]. Therefore, the frequency response from the n th sample in the m th aspect-frame of the target d (here $n \in \{1, 2, \dots, N\}$ with N denoting the number of samples in each aspect-frame, $m \in \{1, 2, \dots, M_d\}$ with M_d denoting the number of aspect-frames of the d th target, and $d \in \{1, 2, \dots, D\}$ with D denoting the number of targets) can be represented by

$$\mathbf{y}_n^{(d,m)} = \Phi \mathbf{w}_n^{(d,m)} + \boldsymbol{\eta}_n^{(d,m)}, \Phi = [\phi(r_1), \phi(r_2), \dots, \phi(r_K)]$$

$$\phi(r_k) = [1, \exp(-j\frac{4\pi}{c}r_k\Delta f), \dots, \exp(-j\frac{4\pi}{c}r_k(L-1)\Delta f)]^T \quad (1)$$

where $\mathbf{y}_n^{(d,m)} = [y_n^{(d,m)}(1), y_n^{(d,m)}(2), \dots, y_n^{(d,m)}(L)]^T$ with $y_n^{(d,m)}(l)$ representing the l th frequency response from the n th sample in the m th aspect-frame of the target d ($l \in \{1, \dots, L\}$ and L denoting the number of frequency components), $\mathbf{w}_n^{(d,m)} = [w_n^{(d,m)}(1), w_n^{(d,m)}(2), \dots, w_n^{(d,m)}(K)]^T$ denotes the n th HRR complex sample in the m th aspect-frame of the target d with $w_n^{(d,m)}(k)$ representing the scattering coefficient from the k th range cell ($k \in \{1, \dots, K\}$ and K denoting the number of range cells), $\boldsymbol{\eta}_n^{(d,m)} = [\eta_n^{(d,m)}(1), \eta_n^{(d,m)}(2), \dots, \eta_n^{(d,m)}(L)]$ represents the corresponding noise component, c denotes the propagation velocity of radar signal, $\{r_k\}_{k=1}^K$ denotes the locations of range cells, Δf denotes the frequency interval between the neighboring frequency components. When $K = L$, Φ is a complete orthogonal basis matrix; when $K > L$, Φ is redundant basis matrix, and the superresolution representation can be achieved.

Our Bayesian sparse model for HRR complex data from the m th aspect-frame of the d th target with $d = 1, 2, \dots, D$ and $m = 1, 2, \dots, M_d$ may be represented as

$$\mathbf{y}_n^{(d,m)} = \Phi \mathbf{w}_n^{(d,m)} + \boldsymbol{\eta}_n^{(d,m)}, \quad \mathbf{w}_n^{(d,m)} = \hat{\mathbf{w}}_n^{(d,m)} \circ \mathbf{z}_n^{(d,m)}$$

$$z_n^{(d,m)}(k) \sim \text{Ber}(u_{k,n}^{(d,m)})$$

$$u_{k,n}^{(d,m)} \sim \begin{cases} \text{Beta}(e_0, f_0) & \text{if } n = 1 \\ \text{Beta}(e_0^0, f_0^0) & \text{if } n \geq 2, z_{n-1}^{(d,m)}(k) = 0 \\ \text{Beta}(e_0^1, f_0^1) & \text{if } n \geq 2, z_{n-1}^{(d,m)}(k) = 1 \end{cases}$$

$$\hat{\mathbf{w}}_n^{(d,m)} \sim \text{CNorm}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0), \quad \eta_n^{(d,m)}(l) \sim \text{CNorm}(0, \alpha^{-1})$$

$$\alpha \sim \text{Gamma}(a_0, b_0)$$

$$n = 1, 2, \dots, N; k = 1, 2, \dots, K; l = 1, 2, \dots, L; K \geq L \quad (2)$$

In the model, $\text{Ber}(\cdot)$ and $\text{Beta}(\cdot)$ refer respectively to the Bernoulli and Beta densities parameterized by the arguments inside parentheses, and hyperparameters $e_0 > 0$, $f_0 > 0$. This Bernoulli-Beta prior is a finite approximation of the Beta process [15]. Note that $\mathbb{E}[u_{k,1}^{(d,m)}] = e_0/(e_0 + f_0)$, where $\mathbb{E}[\cdot]$ denotes the expectation of a random variable. As discussed in [16] for CS-based ISAR imaging, the target signal can be approximated by the echoes from the dominant scattering centers which are assumed to be sparsely distributed in the target, while those from the weak scattering centers are regarded as the noise components. Therefore, to impose sparseness on $\mathbf{z}_1^{(d,m)}$, we may let $e_0/(e_0 + f_0) \approx 0$, and

therefore it is probable that $u_{k,1}^{(d,m)}$ is close to zero. Since K is usually a large integer (especially for the superresolution case), we specify $e_0 = 1/K$ and $f_0 = (K - 1)/K$ in our experiments.

Considering the temporal dependency between HRR samples from a given aspect-frame, if there is a dominant scattering center in a specific range cell in a sample, it is very likely that this dominant scattering center will appear in about the same range cell in the next sample. Here we assume the temporal relationship across $\{\mathbf{z}_n^{(d,m)}\}_{n=2}^N$ satisfies a simple Markov structure, and let $e_0^1 = f_0^0 = 0.9 \times K$ and $f_0^1 = e_0^0 = 0.1 \times K$.

$\text{CNorm}(\cdot)$ represents the complex Gaussian distribution with $p(\hat{\mathbf{w}}_n^{(d,m)} | \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) = \frac{1}{(\pi)^K |\boldsymbol{\Sigma}_0|} \exp\{-(\hat{\mathbf{w}}_n^{(d,m)} - \boldsymbol{\mu}_0)^H \boldsymbol{\Sigma}_0^{-1} (\hat{\mathbf{w}}_n^{(d,m)} - \boldsymbol{\mu}_0)\}$. We set the prior parameters $\boldsymbol{\mu}_0 = \mathbf{0}$ and $\boldsymbol{\Sigma}_0 = \mathbf{I}_K$ for $\hat{\mathbf{w}}_n^{(d,m)}$, i.e., we use the complex Gaussian distribution with the zero-mean and identity-covariance as the prior for $\hat{\mathbf{w}}_n^{(d,m)}$. The posterior mean and covariance parameters can be further inferred for each sample. We assume the model noise is drawn *i.i.d.* from a complex Gaussian distribution with the zero-mean, i.e., $p(\eta_n^{(d,m)}(l) | \alpha) = (\pi\alpha^{-1})^{-1} \exp(-\alpha |\eta_n^{(d,m)}(l)|^2)$. The noise variance or precision is assumed unknown, and is learned within the model inference. $\text{Gamma}(\cdot)$ represents the Gamma distribution with hyperparameters $a_0 > 0$ and $b_0 > 0$. We usually set $a_0 = b_0 = 10^{-6}$ [17].

The posterior distributions of the model parameters Ψ are inferred from the observed training data $\mathbf{Y}$, via the VB inference [18], which converges fast and is computationally efficient (by contrast, Gibbs-sampling [19] convergence may be difficult to assess). In VB inference, we seek a set of distributions $q(\Psi) = \prod_h q_h(\Psi_h)$ to approximate the true posterior distributions $p(\Psi | \mathbf{Y})$, and use a lower bound $\mathcal{L}(q(\Psi))$ to approximate the true log-likelihood of the model $\log p(\mathbf{Y} | \Psi)$. The algorithm iteratively updates $q(\Psi)$ so that $\mathcal{L}(q(\Psi))$ approaches to $\log p(\mathbf{Y} | \Psi)$ until convergence.

III. STATISTICAL RECOGNITION

In Section II, we present a new signal model for the HRR complex data. Although the temporal dependency across samples from a given aspect-frame is exploited, this model mainly describes the scattering property of a single sample. However, for the statistical recognition, based on such a signal model, we should further derive the statistical model for a set of samples from each aspect-frame, which is referred to as statistical frame model.

By statistical recognition is meant that the class membership of a measurement frequency response of an HRR complex sample $\mathbf{y}^*$ is determined based on the posterior probabilities of all class memberships $p(d | \mathbf{y}^*)$ ($d \in \{1, 2, \dots, D\}$ denotes class membership). According to Bayes theorem, we have $p(d | \mathbf{y}^*) \propto p(\mathbf{y}^* | d)p(d)$, where $p(\mathbf{y}^* | d)$ is the class-conditional likelihood and $p(d)$ is the prior class probability. Since the prior class probability is usually assumed to be uniformly distributed, estimation of the posterior probability

is turned into estimation of the class-conditional likelihood. Furthermore, considering the target-aspect sensitivity of HRR data, we need to develop statistical model for each aspect-frame, and the class-conditional likelihood of each class will be obtained via substituting the test sample into all of the target-aspect dependent statistical models and searching the maximum one, *i.e.*, $p(\mathbf{y}^*|d) = \max_m p(\mathbf{y}^*|d, m)$ with $p(\mathbf{y}^*|d, m)$ denoting the pdf of the m th frame from the d th target and $m \in \{1, 2, \dots, M_d\}$. Therefore, the main task in the training stage of statistical recognition is to estimate the statistical model, *i.e.*, pdf, for the data from each aspect-frame of each target.

A. Training stage

After the signal model is inferred for each sample from a given aspect-frame, the corresponding statistical frame model can be estimated as follows

$$p(\mathbf{y}|d, m) = \int \int \text{CNorm}(\mathbf{y}; \Phi \mathbf{w}^{(d, m)}, \alpha^{-1} \mathbf{I}_L) \times \text{Gamma}(\alpha) \times p(\mathbf{w}^{(d, m)}) d\alpha d\mathbf{w} \\ m = 1, 2, \dots, M_d; d = 1, 2, \dots, D \quad (3)$$

where $\text{Gamma}(\alpha)$ represents the posterior Gamma distribution of α after the signal model inference, $p(\mathbf{w}^{(d, m)})$ denotes the posterior pdf for all HRR complex samples from the m th frame of the d th target.

According to the signal model in (2), for each sample in the aspect-frame, $\mathbf{w}_n^{(d, m)} = \hat{\mathbf{w}}_n^{(d, m)} \circ \mathbf{z}_n^{(d, m)}$ with $\hat{\mathbf{w}}_n^{(d, m)} \sim \text{CNorm}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$ and $\{\mathbf{z}_n^{(d, m)}\}_{n=1}^N$ imposed a Markov structure. Since the posterior distribution of $\hat{\mathbf{w}}_n^{(d, m)}$ depends on the posterior mean of $\mathbf{z}_n^{(d, m)}$, which is binary when the inference converges, thus the posterior distribution $p(\mathbf{w}_n^{(d, m)})$ approximately equals to the posterior distribution $p(\hat{\mathbf{w}}_n^{(d, m)})$. Since the scattering coefficient vector of each sample in the aspect-frame follows a complex Gaussian distribution, we can assume that the posterior pdf $p(\mathbf{w}^{(d, m)})$ is also a complex Gaussian distribution.

$$\mathbf{w}^{(d, m)} \sim \text{CNorm}(\boldsymbol{\mu}^{(d, m)}, \boldsymbol{\Sigma}^{(d, m)}) \\ \boldsymbol{\mu}^{(d, m)} = \frac{1}{N} \sum_{n=1}^N \mathbb{E}[\hat{\mathbf{w}}_n^{(d, m)}] \\ \boldsymbol{\Sigma}^{(d, m)} = \frac{1}{N} \sum_{n=1}^N \left\{ \begin{array}{l} \left(\mathbb{E}[\hat{\mathbf{w}}_n^{(d, m)}] - \boldsymbol{\mu}^{(d, m)} \right) \\ \cdot \left(\mathbb{E}[\hat{\mathbf{w}}_n^{(d, m)}] - \boldsymbol{\mu}^{(d, m)} \right)^H \end{array} \right\} \quad (4)$$

where $\mathbb{E}[\hat{\mathbf{w}}_n^{(d, m)}]$ represents the posterior expectation of $\hat{\mathbf{w}}_n^{(d, m)}$. Paper [20] used the similar estimator from Gibbs sampling for deriving the CS inversion.

Since the phase term of the k th dominant scatterer in the n th sample $\theta_k(n) = -\frac{4\pi}{\lambda} r_k(n)$, where λ denotes the radar wavelength and $r_k(n)$ denotes the radial distance from this scatterer to the radar, its phase variation within an aspect-frame $\Delta\theta_k = -\frac{4\pi}{\lambda} \Delta r_k$ is strongly sensitive to the target position variation, where Δr_k denotes the radial distance variation of this scatterer within this aspect-frame. For example,

for the carrier frequency of 5520 MHz (approximately, the wavelength of 5 cm), when $\Delta r_k = 2.5 \text{ cm}$, $\Delta\theta_k = 2\pi$. Thus if we assume that the radial distances of the most scatterers in a target $\{r_k(n)\}_k$ uniformly vary and the corresponding phase terms $\{\theta_k(n)\}_k$ are uniform-distributed within $[0, 2\pi)$ across the samples within the aspect-frame (*i.e.*, for $n = 1, 2, \dots, N$), then the mean value of the complex echoes from each of these scatterers within the aspect-frame must approximate zero. In this sense, it is reasonable to assume that $\mathbf{w}^{(d, m)}$ follows a complex Gaussian distribution with zero-mean. Therefore, for our application, (4) can be simplified as

$$\mathbf{w}^{(d, m)} \sim \text{CNorm}(\mathbf{0}, \boldsymbol{\Sigma}^{(d, m)}) \\ \boldsymbol{\Sigma}^{(d, m)} = \frac{1}{N} \sum_{n=1}^N \mathbb{E}[\hat{\mathbf{w}}_n^{(d, m)}] \mathbb{E}[\hat{\mathbf{w}}_n^{(d, m)}]^H \quad (5)$$

Then due to the conjugacy, we can calculate the integrals in (3) and get the frame model

$$p(\mathbf{y}|d, m) = \text{CNorm} \left(\mathbf{0}, \begin{array}{l} \Phi \left(\frac{1}{N} \sum_{n=1}^N \mathbb{E}[\hat{\mathbf{w}}_n^{(d, m)}] \mathbb{E}[\hat{\mathbf{w}}_n^{(d, m)}]^H \right) \Phi^H \\ + \left(\mathbb{E}[\alpha] \right)^{-1} \mathbf{I}_L \end{array} \right) \\ m = 1, 2, \dots, M_d; d = 1, 2, \dots, D \quad (6)$$

where the posterior expectations $\{\mathbb{E}[\hat{\mathbf{w}}_n^{(d, m)}]\}_{n=1}^N$ and $\mathbb{E}[\alpha]$ can be easily got after the signal model inference.

B. Classification stage

If the frequency response of a test HRR complex sample, *i.e.*, $\mathbf{y}^*$, is measured under the same SNR circumstance as that of training data, its frame-conditional likelihood given the frame m of the target d can be calculated according to (6). Based on the property of complex Gaussian distribution,

$$p(\mathbf{y}^*|d, m) = \text{CNorm}(\mathbf{y}^*; \mathbf{0}, \boldsymbol{\Lambda}^{(d, m)}) \\ \boldsymbol{\Lambda}^{(d, m)} = \begin{array}{l} \Phi \left(\frac{1}{N} \sum_{n=1}^N \mathbb{E}[\hat{\mathbf{w}}_n^{(d, m)}] \mathbb{E}[\hat{\mathbf{w}}_n^{(d, m)}]^H \right) \Phi^H \\ + \left(\mathbb{E}[\alpha] \right)^{-1} \mathbf{I}_L \end{array} \quad (7)$$

Since the frame models from a given target are assumed to be independent [4], then the class-conditional likelihood of target d ($d \in \{1, 2, \dots, D\}$) is $p(\mathbf{y}^*|d) = \max_m p(\mathbf{y}^*|d, m)$ with $m \in \{1, 2, \dots, M_d\}$. According to Bayes theorem $p(d|\mathbf{y}^*) = \frac{p(\mathbf{y}^*|d)p(d)}{\sum_{d=1}^D p(\mathbf{y}^*|d)p(d)}$ with $d = 1, 2, \dots, D$, the class membership of $\mathbf{y}^*$ can be assigned to the class with the maximum posterior probability, *i.e.*, $d^* = \arg \max_d p(d|\mathbf{y}^*)$. Similar frame matching method is also utilized in [4], [5], [7], [8].

However, for the RATR problem, the training data can usually be obtained under the high SNR via some cooperative measurement experiments or directly via simulations; while the test samples are usually got in the non-cooperative circumstance (*e.g.*, at the battle time), where the high SNR cannot be guaranteed due to the large distance between the non-cooperative targets and radar. If we assume that the test sample measured under the low SNR can be expressed as

$\tilde{\mathbf{y}}^* = \mathbf{y}^* + \boldsymbol{\nu}^*$, where the noise level of $\mathbf{y}^*$ is same as those of the training samples (namely with the high SNR) and $\boldsymbol{\nu}^*$ is the new noise component introduced from the test condition with the assumption that $\boldsymbol{\nu}^* \sim \text{CNorm}(\mathbf{0}, r_*^{-1} \mathbf{I}_L)$, we can have

$$\begin{aligned} p(\tilde{\mathbf{y}}^*|d, m) &= \text{CNorm}(\tilde{\mathbf{y}}^*; \mathbf{0}, \tilde{\boldsymbol{\Lambda}}^{(d, m)}) \\ \tilde{\boldsymbol{\Lambda}}^{(d, m)} &= \boldsymbol{\Lambda}^{(d, m)} + r_*^{-1} \mathbf{I}_L \end{aligned} \quad (8)$$

Therefore, if we can estimate the test noise level r_* via some pre-processing methods, the matched target indicator c^* and frame indicator m^* can be easily obtained via the above frame matching method based on $\{p(\tilde{\mathbf{y}}^*|d, m)\}_{m=1, d=1}^{M_d, D}$. The noise considered here mainly refers to the heat noise from the radar receiver, which can be assumed to be stationary, accordingly, there are several ways to accurately estimate the noise power (*i.e.*, noise variance r_*^{-1}) [21], of which a simple approach is to measure the noise power from the radar receiver with no input signal.

In this subsection, we mainly discuss how to update the statistical frame model trained under the high SNR to match the test data collected under the low SNR. The frame matching method discussed above is also used as the classification method.

IV. EXPERIMENTAL RESULTS

We examine the performance of the proposed model on the measured airplane data used in [4], [5], [8]. The center frequency and bandwidth of the radar are 5520MHz and 400MHz . The projections of target trajectories onto the ground plane are shown in Figure 1. In order to test the generalization performance of the recognition methods, the 2nd and the 5th segments of Yak-42, the 6th and the 7th segments of Cessna, the 5th and the 6th segments of An-26 are taken as the training samples, other data segments are taken as test samples in our experiments. Same as those in our previous work [4], [5], [8], [13], we use the aspect-frame partition method to deal with the target-aspect sensitivity of the training dataset. There are 35 aspect-frames for Yark-42, 50 for Cessna Citation S/II, and 50 for An-26, of which each aspect-frame totally has 1024 HRR samples, *i.e.*, $D = 3$, $M_1 = 35$, $M_2 = M_3 = 50$, $N = 1024$.

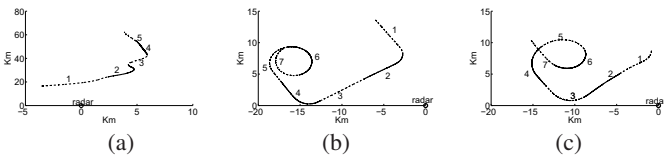


Fig. 1. Projections of target trajectories onto the ground plane: (a) Yak-42; (b) Cessna Citation S/II; (c) An-26.

To illustrate the accuracy of the parameter estimation of our proposed Bayesian sparse model, a simulation experiment is performed in the following. The range resolution of our measured data is around 0.375m , since $\Delta R \approx \frac{c}{2B}$ with ΔR denoting the range resolution, $c = 3.0 \times 10^8\text{m/s}$ denoting the propagation velocity of radar signal and B denoting the signal bandwidth. In the following experiment, the range resolution

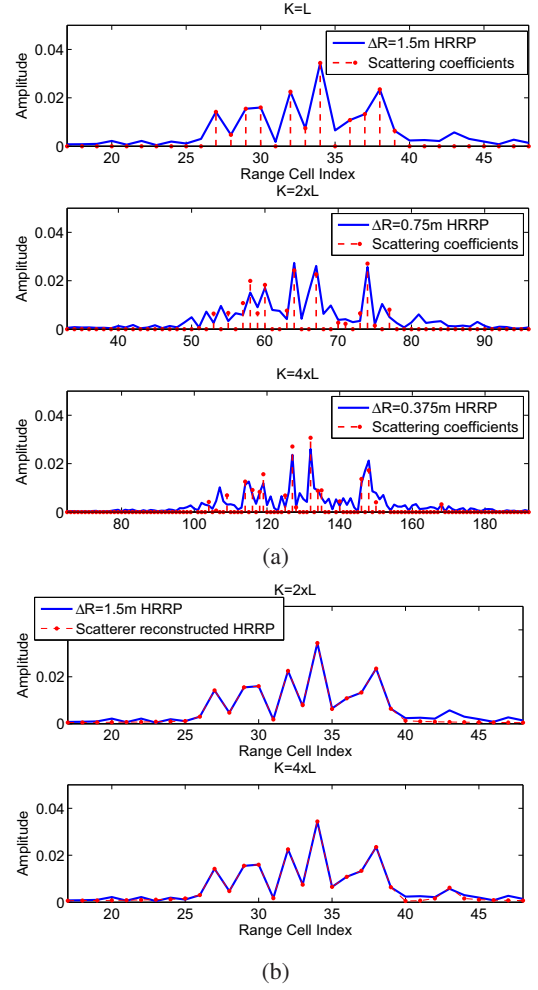


Fig. 2. A signal model learning example from Yak-42 HRR data. The signal model learning and reconstructions use 1.5m resolution data, *i.e.*, $L = 64$. (a) First row: Reference profile at 1.5m resolution, and the learned amplitudes of the scattering coefficients via the Bayesian sparse model with $K = L$ (*i.e.*, $\{|w_n^{(d, m)}(k)|\}_{k=1}^{64}$ and $\Delta R = 1.5\text{m}$); Second row: Reference profile at 0.75m resolution, and the learned amplitudes of the scattering coefficients via the Bayesian sparse model with $K = 2 \times L$ (*i.e.*, $\{|w_n^{(d, m)}(k)|\}_{k=1}^{128}$ and $\Delta R = 0.75\text{m}$); Third row: Reference profile at 0.375m resolution, and the learned amplitudes of the scattering coefficients via the Bayesian sparse model with $K = 4 \times L$ (*i.e.*, $\{|w_n^{(d, m)}(k)|\}_{k=1}^{256}$ and $\Delta R = 0.375\text{m}$). (b) First row: Reference profile at 1.5m resolution, and the reconstructed 1.5m HRRP via the Bayesian sparse model with $K = 2 \times L$; First row: Reference profile at 1.5m resolution, and the reconstructed 1.5m HRRP via the Bayesian sparse model with $K = 4 \times L$.

was decreased from 0.375m to 1.5m (*i.e.*, $L = 64$) to test our signal model, and the 0.375m (*i.e.*, $L = 256$) or 0.75m (*i.e.*, $L = 128$) HRRP samples were used as the “reference” profiles. Figure 2 shows some examples from Yak-42 data. As shown in the first subfigure in Figure 2 (a), our signal model can learn the locations of the dominant scatterers, and their learned amplitudes are exactly same as those in the 1.5m HRRP, since Φ is a complete orthogonal basis matrix with $K = L$. For the superresolution cases, *i.e.*, the second and third subfigures in Figure 2 (a), since Φ is nonorthogonal and some real scattering centers in the target may not be at the locations

corresponding to the bases in Φ , the learned amplitudes of the scattering coefficients are not exactly same as those in the $0.75m$ HRRP or $0.375m$ HRRP. Nevertheless, our signal model finds the most dominant scatterers located at the bases, and the learned profiles approximate to the reference $0.75m$ HRRP or $0.375m$ HRRP. Figure 2 (b) gives the reconstructed $1.5m$ HRRPs via our Bayesian sparse model with $K = 2 \times L$ and $K = 4 \times L$, which are very similar to the reference $1.5m$ HRRP. The larger K is, the more precise the reconstruction is. Figure 2 (b) further demonstrates the accuracy of the parameter estimation of our signal model.

Figure 3 shows $\{\mathbb{E}[z_n^{(2,1)}]\}_{n=1}^{1024}$ for the $0.375m$ HRR data (i.e., $L = 256$) from the first aspect-frame of Cessna Citation S/II, which are learned via the Bayesian sparse model with $K = L$, $K = 2 \times L$ and $K = 4 \times L$. For the m th aspect-frame of the d th target, if a dominant scatterer is located at the n th range cell, then the posterior probability on this range cell usage, $\mathbb{E}[z_n^{(d,m)}]$, should be one; otherwise, it will be zero. Each column in each subfigure of Figure 3 denotes the inferred locations of the dominant scatterers for each sample in the frame. As shown in Figure 3, the larger K is, the more precise superresolution we can achieve. Given the parameter K , most dominant scatterers from the temporally proximate HRR samples are located at the same range cells, and also the distant repetitions or “innovations” occur for some samples, which accords with our Markov structure assumption. In addition, the number of the dominant scattering centers of an HRR sample varies across the sample index, which justifies using the Beta process prior for the automatic model selection.

This HRR dataset is measured under the high SNR. For aircraft-like targets under a look-up scenario such as the case of this dataset, the noises in the inphase and quadrature echoes of targets can be assumed to be Gaussian white noises. Therefore, in order to evaluate the noise robustness of our proposed methods, we add simulated white noises to the inphase and quadrature components of the complex frequency responses from the HRR complex test samples. The SNR is defined as

$$\text{SNR} = 10 \times \log_{10} \left(\frac{\bar{P}_y}{P_{\text{Noise}}} \right) = 10 \times \log_{10} \left(\frac{\sum_{l=1}^L P_{y_l}}{L \times P_{\text{Noise}}} \right) \quad (9)$$

where $\bar{P}_y$ denotes the average power of the complex frequency response from an original HRR complex test sample, P_{y_l} denotes the power of the l th frequency response from the original test sample, L denotes the number of frequency components (here $L = 256$), and P_{Noise} denotes the power of noise in each frequency component. Thus $10^{\frac{\text{SNR}}{10}}$ equals to the rate of the average power of original HRR sample to the power of the noise.

Figure 4 depicts the average recognition rates versus SNR, generated via the five noise-robust statistical recognition methods, including the two versions of the noise-robust MTL-FA model [8], i.e., the noise-robust MTL-FA model with the test noise level prior and the noise-robust MTL-FA model with the adaptively learned test noise level, and the robust recognition

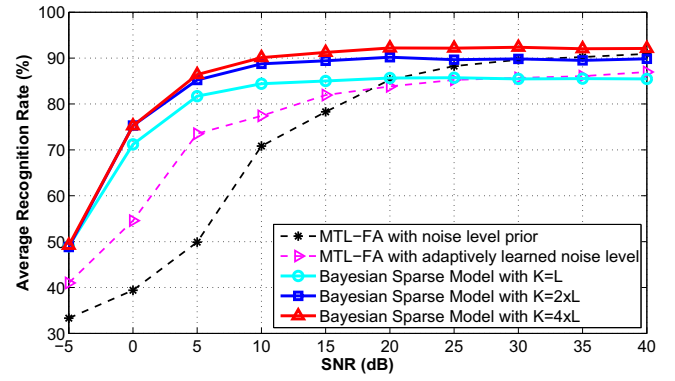


Fig. 4. Variation of the recognition performance with SNR, via the noise-robust MTL-FA model with the test noise level prior [8], the noise-robust MTL-FA model with the adaptively learned test noise level [8], the robust recognition methods based on Bayesian sparse models with $K = L$, $K = 2 \times L$ and $K = 4 \times L$, where $K > L$ means the superresolution representations for the HRR complex data. All these methods are trained with the training data under the high SNR, but the classifications are implemented under different SNR conditions. Here the two versions of the MTL-FA model are used for the FFT-magnitude feature extracted from the HRR complex data, and the three versions of the Bayesian sparse model are directly for the complex frequency responses from the HRR complex data. The recognition rates are averaged over ten runs.

methods based on our Bayesian sparse models with $K = L$, $K = 2 \times L$ and $K = 4 \times L$. All these methods are trained with the training data under the high SNR, but the classifications are implemented under different SNR conditions. Here the two MTL-FA models are used for the FFT-magnitude feature extracted from the HRR complex data, and the three Bayesian sparse models are directly for the complex frequency responses from the HRR complex data. For brevity in Figure 4, we omit the recognition curves from the six statistical models that are non-robust to noise, i.e., Gaussian model [1], [2], Gamma model [3], the two-distribution compounded model [4], the single task learning based FA (STL-FA) model [5], the dynamic MTL-HMM model [7], and the MTL-FA model without modification in classification [8]. From [7], [8], the two noise-robust MTL-FA models obviously outperform the six non-robust models under the condition of low SNR.

As shown in Figure 4, our Bayesian sparse model with $K = 4 \times L$ is superior to other four models under each SNR case; when $\text{SNR} \geq 35\text{dB}$, the noise-robust MTL-FA model with the test noise level prior yields a little better recognition rates than our Bayesian sparse model with $K = 2 \times L$, and the noise-robust MTL-FA model with the adaptively learned test noise level yields a little better recognition rates than our Bayesian sparse model with $K = L$; when $\text{SNR} \leq 15\text{dB}$, the three Bayesian sparse models outperform the two noise-robust MTL-FA models; when $\text{SNR} \geq 5\text{dB}$, the recognition rates of our new robust recognition methods change slightly, which demonstrates the statistical models updated via (8) and the noised complex test data match well in the classification stage. If we assume that the average recognition rate threshold for the real application is 70%, the noise-robust recognition methods based on our Bayesian sparse models can work under the

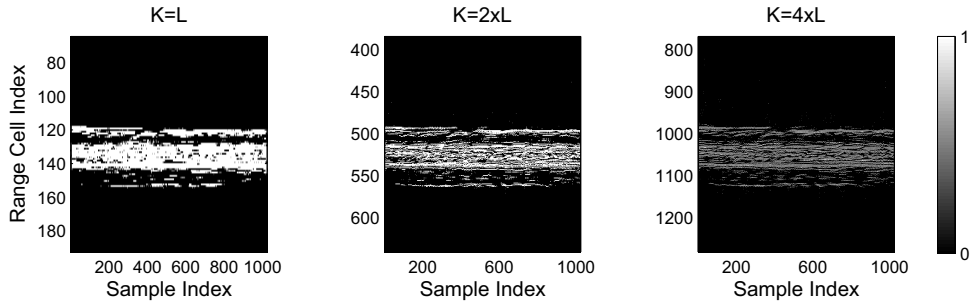


Fig. 3. Learned locations of the dominant scatterers for the 0.375m HRR data (i.e., $L = 256$) from the first aspect-frame of Cessna Citation S/II, i.e., $\mathbf{Z}^{(2,1)} = [\mathbb{E}[\mathbf{z}_1^{(2,1)}], \mathbb{E}[\mathbf{z}_2^{(2,1)}], \dots, \mathbb{E}[\mathbf{z}_{1024}^{(2,1)}]]$, generated via the Bayesian sparse models with $K = L$ (left subfigure), $K = 2 \times L$ (middle subfigure) and $K = 4 \times L$ (right subfigure).

test condition of $\text{SNR} \geq 0\text{dB}$; while the noise-robust MTL-FA model with the test noise level prior and the noise-robust MTL-FA model with the adaptively learned test noise level need $\text{SNR} \geq 10\text{dB}$ and $\text{SNR} \geq 5\text{dB}$. Thus, according to the radar equation, the 10dB or 5dB advantage in SNR via our new methods can bring an increase about 80% or 35% in the classification distance between the target and radar in the real application.

V. CONCLUSION

We have developed a Bayesian sparse model for HRR complex data, where a sparseness-promoting prior on the complex scattering coefficients and a Markov dependency for the location of the dominant scattering center between consecutive HRR signals are imposed. Based on such a signal model, the statistical frame model for the HRR complex data from each target-aspect is analytically derived, where the statistical frame model trained under the high SNR can be updated to match the test sample measured under the low SNR and the corresponding recognition decision can be made based on the updated templates. The RATR experiments based on measured HRR data show that the proposed method has better recognition under the test condition of the low SNR than the existing statistical methods for HRR data.

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